	Course	**OPRE 3360.005: Managerial Methods in Decision Making Under Uncertainty**
	Professor	A. Serdar Simsek
	Term	Spring 2026
	Meetings	Tuesday & Thursday 2:30pm – 3:45pm, JSOM 1.212

PROFESSOR'S CONTACT INFORMATION

Office Location	JSOM 3.227
Email Address	serdar.simsek@utdallas.edu
Office Hours	Tuesday 4:30pm – 5:30pm or by appointment
Other Information	Please include "OPRE 3360" in the title of your email to ensure my timely response.
TA	Shenghui Zhai, Shenghui.Zhai@UTDallas.edu. Office hours TBA.

GENERAL COURSE INFORMATION

Pre-Requisites	MATH 2413 or MATH 2417.
Other Restrictions	Credit cannot be received for both OPRE 3360 and STAT 3360 courses.
Course Description	This course focuses on the concepts of probability and statistics for managerial decision-making. Concepts will be developed in lectures and exercises using software packages and topics including: summarizing and presenting data, probability theory, sampling, estimation, confidence intervals, hypothesis testing, regression, and ANOVA. Emphasis will be given to modeling and solving business problems in finance, marketing, accounting and operations management.
Learning Outcomes	On successful completion of this course a student will be able to use the tools of probability and statistical modeling to support business decisions. Specifically, she/he (i) can quantify uncertainty found in business situations and form probabilistic knowledge, (ii) estimate the key metrics based on data and test hypotheses about business realities using data, and (iii) build a statistical model of uncertain business realities for insight gathering and prediction.
Required Texts & Materials	Modern Business Statistics with Microsoft Excel (8th edition) by Camm, Cochran, Fry, Ohlmann, Anderson, Sweeney, Williams (with MindTap) You will need to purchase at least the digital version of this required text because your individual assignments will be completed on **MindTap**. **Please use the link on eLearning to purchase the required text.** Here is a video on how to register and access the MindTap course: How to Register for MindTap in eLearning. (The video refers to "Blackboard," which is the platform our school calls "eLearning.")

Other Texts, Readings & Materials

Other materials and lecture notes will be posted on eLearning.

If students would like to read another textbook, the following is recommended:

- (Optional) Groebner, D., Shannon, P. and Fry, P. "Business Statistics: A Decision-Making Approach," 10th ed.

The required textbook, lecture notes, and other materials posted on eLearning should be sufficient for the students to learn the material. This optional textbook is not required.

COURSE POLICIES

Grading (credit) Criteria

The total score (100 points) will be made up of five parts.

- **Individual assignments**
  - **MindTap assignments (30%).** Most individual assignments are **due by 11:59 PM on the day of the Tuesday lecture** and should be completed via the MindTap system. You can attempt each question up to three times (only your best attempt will be graded) and will receive immediate grading and feedback upon submission. Some questions may include algorithmically generated slight variations across students. All MindTap assignments carry **equal weight** in your overall grade, and **the lowest assignment grade will be dropped** at the end of the semester.
  - **Assessment assignment (6%).** A single homework assignment that will be posted on eLearning during Week 11.
- **TEST I (27%).**
- **TEST II (33%).**
- **Class participation (4%).** Class attendance is highly encouraged. If you must miss a class because of an emergency you will have to contact the professor in advance for permission and provide proof, and you may need to meet with the professor individually to make up the class. You are also expected to actively participate in class. Active participation includes asking questions about the lecture content when it is not clear to you and answering the questions the professor asks in class. I will occasionally use cold calls to increase the participation level. I reserve the right to use my discretion in class participation score based on your "active participation." There may also be 5-minute pop quizzes at the **start** of some randomly selected classes. These quizzes will not be graded and are solely intended for recording attendance.

Exam Location and Testing Center Registration

Both tests (Test 1 and Test 2) will be held at Testing Center on the first floor of the Synergy Park North 2 building --SPN2-- (3020 Waterview Parkway). Note that the exam location is NOT the regular classroom.

- Begin registering for **BOTH Test 1 & Test 2** as soon as Spring classes begin (**recommended**), or **no later than 48 hours prior to the exam time** via this link https://ets.utdallas.edu/testing-center/students/
- Review the Student Guidelines prior to taking the exam at the center.
- ARC students with approved accommodations should review the ARC Testing Guidelines prior to taking their exam at the center.

Letter grade You will be evaluated relative to your classmates.

Extra Credit Extra credit will **NOT** be offered.

Late Work & Make-ups

Excused, no penalty (with documentation):

- **Medical emergency** (proof from a credible medical institution).
- **UTD academic activity** (signed letter from a UTD faculty member).

If excused, deadlines are extended as needed with no late penalty.

MindTap assignments:

- Late penalty: **30% of the maximum points per day** (1 day = 30%, 2 days = 60%, 3 days = 90%). After **3 days late → 0**.
- You **must email me** to request access; MindTap closes automatically at the deadline and I have to reopen it.

Assessment assignment:

- Up to **1 day late only**, **35%** penalty (solutions are posted promptly for exam preparation; no submissions after that).

Exams:

- **Documented medical reasons or UTD academic activity:** make-up with **no penalty**.
- **Non-medical misses** (e.g., missing testing-center registration or day-of-exam issues): make-up allowed with a **40%** penalty.

How penalties are applied:
A "X% penalty" means **X% of the assignment's/exam's maximum points** is subtracted from your score.

Technical Requirements In addition to a confident level of computer and Internet literacy, certain minimum technical requirements must be met to enable a successful learning experience. Please review the important technical requirements on the Getting Started with eLearning webpage.

Course Access and Navigation This course can be accessed using your UT Dallas NetID account on the eLearning website.

Please see the course access and navigation section of the Getting Started with eLearning webpage for more information.

To become familiar with the eLearning tool, please see the Student eLearning Tutorials webpage.

UT Dallas provides eLearning technical support 24 hours a day, 7 days a week. The eLearning Support Center includes a toll-free telephone number for immediate assistance (1-866-588-3192), email request service, and an online chat service.

Communication This course utilizes online tools for interaction and communication. Some external communication tools such as regular email and a web conferencing tool may also be used during the semester. For more details, please visit the Student eLearning Tutorials webpage for video demonstrations on eLearning tools. Student emails and discussion board messages will be answered within 3 working days under normal circumstances.

Server Unavailability or Other Technical Difficulties The University is committed to providing a reliable learning management system to all users. However, in the event of any unexpected server outage or any unusual technical difficulty which prevents students from completing a time sensitive assessment activity, the instructor will provide an appropriate accommodation based on the situation. Students should immediately report any problems to the instructor and also contact the online eLearning Help Desk. The instructor and the eLearning Help Desk will work with the student to resolve any issues at the earliest possible time.

Class Participation Regular class participation is expected regardless of course modality. Students who fail to participate in class regularly are inviting scholastic difficulty. A portion of the grade for this course is directly tied to your participation in this class. It also includes engaging in group or other activities during class that solicit your feedback on homework assignments, readings, or materials covered in the lectures (and/or labs). Class participation is documented by faculty. Successful participation is defined as consistently adhering to University requirements, as presented in this syllabus. Failure to comply with these University requirements is a violation of the Student Code of Conduct.

Classroom Citizenship Classes begin on time. Please maintain class decorum and be respectful toward fellow students. If you have a doubt or misunderstanding regarding course work feel free to discuss it with me.

Software The course will involve extensive use of Microsoft Excel, and in particular the data-analysis tool pack (a native Excel add-in). The use of each tool will be discussed in class but basic familiarity (such as copy and paste, entering formulae) with Microsoft Excel is assumed. In case some students may have laptops with earlier versions of MS Office, the lecture notes, classroom demonstrations, and screen shots will utilize features common for all versions of Excel.

Mac Support Microsoft Office for Mac differs from Microsoft Office for Windows in terms of available features, particularly statistical tools. If you are using a Mac and need assistance, you may contact the UTD IT Help Desk for support. Please note that I may not be able to assist with Mac-specific issues, as I am not familiar with the Mac version of Office. Additionally, Exams 1 and 2 will require the use of Microsoft Office for Windows installed in the Testing Center computers. UTD provides support for students purchasing Microsoft Office for Windows: please check with the university technology store.

STRATEGIES FOR SUCCESS

1. **Class preparation:**

   The main topics, reading assignments, and suggested problems will posted on eLearning. In each class you will be expected to:

   a) Complete required readings and skim lecture notes (prior to class) if you have time.
   b) Print a hardcopy of the lecture notes for each class and bring it with you to take additional notes.
   c) Do all required Excel pre-work assignments prior to class and bring a laptop for Module III.
   d) Solve the suggested self-assessment problems in the Practice Problem Set **assigned for each class** (before and after class).
   e) Read the appropriate portions of the textbook (before and after class).

   I understand that you have limited time. However, I strongly urge you to devote adequate time to the course since this is not material you can cram over a day or two. If you are pressed for time for specific classes, prioritize your class preparation in the order given above.

2. **Practice problem sets**

   There are sets of self-assessment practice problems for each lecture starting from Week 2. To maximize your learning, attempt the self-assessment exercises for each lecture soon after the lecture, and review your lecture notes or seek help if you find yourself unable to complete the exercises.

3. **Workload expectation:**

   The key to competence is **PRACTICE**. You will be given the opportunity to work on many practice problems in this course.

   Note that university guidelines recommend that you study 2-3 hours per week for every credit hour in which you are enrolled. University expectations suggest you spend **6-9 hours** outside of class every week on homework and studying for this course.

   A rule of thumb for time requirements is that I expect students will spend approximately 3 hours preparing for and reviewing each class, with additional time spent solving the practice problems and preparing for exams. Trimming this time input will diminish the value of the educational experience for everyone. Please recognize the importance of advance preparation, and begin now to level-load your course time input.

4. **How to get help:**

   In addition to my office hours, a TA will be available for additional office hours for those students who need help. This course is cumulative and moves rapidly. **DO NOT FALL BEHIND**! It is recommended that you see your instructor **immediately for any difficulties.**

TENTATIVE COURSE OUTLINE AND EXAMS CALENDAR

I will attempt to stick to this schedule as much as possible, although I reserve the right to modify the lecture contents and homework due dates depending on the evolution of the course. Sufficient notice will be given for any changes.

Week	Date	Lecture Topics
1	Jan 20-22	Course overview and introduction to data description
2	Jan 27-29	Introduction to probability
3	Feb 3-5	Working with random variables
4	Feb 10-12	Discrete random variables and binomial distribution
5	Feb 17-19	Continuous random variables and normal distribution
	Feb 24	**TEST I – NO REGULAR CLASS MEETING**
6	March 3-5	Sampling distributions and introduction to inference
7	Mar 10-12	Confidence intervals
	Mar 17-19	**NO CLASSES: Spring break**
8	Mar 24-26	Hypothesis test and introduction to statistical models
9	Mar 31- Apr 2	Covariance, correlation, and simple linear regression I
10	Apr 7-9	Simple linear regression II
11	Apr 14-16	Multiple regression and multicollinearity
12	Apr 21-23	Dummy Variables and Applications
13	Apr 28-30	An application of multiple regression and course review
	May 5	**TEST II – NO REGULAR CLASS MEETING**

TENTATIVE COURSE OUTLINE

The following pages list a lecture-by-lecture outline of the entire course. The learning objective, readings, and required and suggested assignments for each lecture are listed. To help you maximize your learning from the lectures and prepare for the two tests, I have assigned a number of individual assignments and practice problems for each class.

I will attempt to stick to this schedule as much as possible, although I reserve the right to modify the lecture contents depending on the evolution of the course. Sufficient notice will be given for any changes.

To help you manage your time efficiently I have categorized the activities on the to-do list into mandatory (√) and optional (▪). I strongly encourage that you complete mandatory activities before or immediately after each class.

Weekly Course Schedules

Module I: Describing and Measuring Data and Uncertainty

Week 1: Course overview and introduction to data description

Learning objectives for this unit:

- Understand the definition and classification of data
- Understand the scale of measurement
- Learn how to summarize data for a categorical variable using graphical methods
- Learn how to summarize data for a quantitative variable using graphical methods

Preparation:

- ✓ Required reading: **super important!** Carefully review the course syllabus and plan!
- ✓ Required reading: read class synopsis, cases, and lecture notes.
- ▪ Text reading: Chap 1: 1.1, 1.2, 1.4–1 .7; Chap 2: 2.1, 2.2 (skip dot plot and stem-and-leaf display).
- ▪ Optional reading or skim: *For Today's Graduate, Just One Word: Statistics*, The New York Times
- ▪ Optional reading or skim: *Are you ready for the era of big data?,* McKinsey Global Institute

After-class:

- ✓ Must-do: read the *Review of Quantitative and Excel Skills* (posted on eLearning).
- ✓ Required assignment: login to MindTap and enter your personal key. Complete the 1st individual assignment before **11:59 PM on Jan 27**.

Week 2: Introduction to probability

Learning objectives for this unit:

- Understand random events, probability spaces, and the basic laws of probability.

Preparation:

- ✓ Required reading: read class synopsis, cases, and lecture notes.
- ▪ Text reading: Chap 2: 2.3; Chap 4: 4.1 (skip the combination and permutation), 4.2, 4.3, 4.4.

After-class:

- ✓ Required assignment: login to MindTap and enter your personal key (if you have not done so). Complete the 2nd individual assignment in MindTap before **11:59 PM on Feb 3**.
- ✓ Must-do: Practice Problem Set 1: 1, 2, 3, 4, 5, 6

Week 3: Working with random variables

Learning objectives for this unit:

- Understand the concept of a random variable and its probability distribution.

Preparation:

- ✓ Required reading: read class synopsis, cases, and lecture notes.
- ▪ Text reading: Chapter 5: 5.1, 5.2 (skip discrete uniform distribution), 5.3, 5.4 (skip Covariance and Correlation for now and focus on cases where the random variables are independent. When random variables X and Y are independent, covariance $\sigma_{xy} = 0$ and correlation $\rho_{xy} = 0$. We will come back to discuss covariance and correlation in Module III).

After-class:

- ✓ Required assignment: complete the 3rd individual assignment in MindTap before **11:59 PM on Feb 10**.
- ✓ Must-do: Practice Problem Set 1: 9, 10, 11, 19–27.

Week 4: Discrete random variables and binomial distribution

Learning objectives for this unit:

- Learn how binomial distribution applies in business decision problems.
- Learn the difference between discrete and continuous random variables.

Preparation:

- ✓ Required reading: read class synopsis, cases, and lecture notes.
- ▪ Text reading: Chapter 5: 5.5.

After-class:

- ✓ Required assignment: complete the 4th individual assignment in MindTap before **11:59 PM on Feb 17**.
- ✓ Must-do: Practice Problem Set 1: 7, 8, 12–18.

Week 5: Continuous random variables and normal distribution.

Learning objectives for this unit:

- Learn how normal distribution applies in business decision problems.
- Preview the Central Limit Theorem.
- Learn types of data available and how to collect data.

Preparation:

- ✓ Required reading: read class synopsis, cases, and lecture notes.
- ▪ Text reading: Chapter 6: 6.2.

After-class:

- ✓ Required assignment: complete the 5th individual assignment in MindTap before **11:59 PM on Monday, Feb 23 (note the different day-of-week than the usual deadlines)**.
- ✓ Must-do: Practice Problem Set 1: 28-34.

Review Session for TEST I: A review session will be provided at the regular class meeting time on Thursday, February 19.

The test will be held at Testing Center at the first floor of the Synergy Park North 2 building --SPN2-- (3020 Waterview Parkway) on Tuesday, February 24 in any 2 hour 30 minutes time period from 8:30am to 9pm. Note that the exam location is NOT the regular classroom. Registration for seat reservation closes 48 hours prior to exam date!

TEST I covers the first course module. For preparation, review lecture notes, all your individual assignments, and solve Practice Test I.

Module II: Statistical Inference and Hypothesis Testing

Week 6: Sampling distributions and introduction to inference

Learning objectives for this unit:

- Explore why sampling leads to errors.
- Learn the concept of sampling error and its distribution.

Preparation:

- ✓ Required reading: read class synopsis, cases, and lecture notes.
- ▪ Text reading: Chap 3: 3.1 (skip weighted mean and geometric mean), 3.2. Chap 7: 7.1–7.6.
- ▪ Optional reading or skim: article from The New York Times: *Risk Mismanagement: What led to the financial meltdown.*

After-class:

- ✓ Required assignment: complete the 6th individual assignment in MindTap before **11:59 PM on March 10**.
- ✓ Must-do: Practice Problem Set 2: 1, 2, 3, 4, 5, 6, 7.

Week 7: Confidence interval and hypothesis test

Learning objectives for this unit:

- Learn to construct a confidence interval for a population parameter (mean or proportion).
- Learn how to formulate null and alternative hypotheses.
- Understand the p-value for a hypothesis test and how to compute it.

Preparation:

- ✓ Required reading: read class synopsis, cases, and lecture notes.
- ▪ Text reading: Chap 8: 8.1–8.4.

After-class:

- ✓ Required assignment: complete the 7th individual assignment in MindTap before **11:59 PM on March 24**.
- ✓ Must-do: Practice Problem Set 2: 8, 9, 11, 12, 14.

Week 8: Hypothesis test and introduction to statistical models

Learning objectives for this unit:

- Apply hypothesis test in business decision making.
- Understand the need and use of statistical models.
- Learn how to perform preliminary analysis.

Preparation:

- ✓ Required reading: read class synopsis, cases, and lecture notes.
- ▪ Text reading: Chap 9: 9.1–9.4.

After-class:

- ✓ Required assignment: complete the 8th individual assignment in MindTap before **11:59 PM on March 31**.
- ✓ Must-do: Practice Problem Set 2: 15, 16, 17, 18, 20, 25, 26, 27, 28, 29, 30.

Module III: From Data to Insights: Statistical Modeling and Regression Analysis

Week 9: Covariance, correlation, and simple linear regression

Learning objectives for this unit:

- Be able to establish the relationship between two variables.
- Learn the simple linear regression model.

Preparation:

- ✓ Required reading: read class synopsis and mini-cases.
- ✓ Must-do: download the data files to your laptop.
- ▪ Text reading: Chap 2: 2.4 (scatter diagram and trendline); Chap 3: 3.5; Chap 5: 5.4; Chap 14: 14.1–14.2, 14.7 (use Excel for regression).

After-class:

- ✓ Required assignment: complete the 9th individual assignment in MindTap before **11:59 PM on April 7**.
- ✓ Must-do: Practice Problem Set 3: 1, 4, 5, 6, 7, 8, 16, 19, 20.

Week 10: Simple linear regression II

Learning objectives for this unit:

- Understand how regression can be used to estimate population parameters.
- Determine whether a regression model is significant using sample data.
- Learn how the regression can be applied to estimate the beta coefficient of a stock.

Preparation:

- ✓ Required reading: read class synopsis and mini-cases.
- ✓ Must-do: download the data files to your laptop.
- ▪ Text reading: Chap 14: 14.3–14.8 (skip Using Excel to Compute the Coefficient of Determination, Standard Residuals, Using Excel to Construct a Residual Plot, and Normal Probability Plot).

After-class:

- ✓ Required assignment: complete the 10th individual assignment in MindTap before **11:59 PM on April 14**.
- ✓ Must-do: Practice Problem Set 3: 2, 3, 9, 10, 11, 12, 13, 14.

Week 11: Multiple regression and multicollinearity

Learning objectives for this unit:

- Apply multiple regression analysis to business applications.
- Understand the causes, symptoms, and remedies for multicollinearity.

Preparation:

- ✓ Required reading: read class synopsis and mini-cases.
- ✓ Must-do: download the data files to your laptop.
- ▪ Text reading: Chap 15: 15.1–15.6, 15.8.

After-class:

- ✓ Required assignment: complete the 11th individual assignment in MindTap before **11:59 PM on April 21**.

Week 12: Dummy Variables and Applications

Learning objectives for this unit:

- Understand the use of dummy variables.
- Apply multiple regression analysis with dummy variables to business applications.

Preparation:

- ✓ Required reading: read class synopsis and mini-cases.
- ✓ Must-do: download the data files to your laptop.
- ▪ Text reading: Chap 15: 15.7.

After-class:

- ✓ Required assignment: **complete the assessment assignment before 11:59 PM on May 1**. For instructions please see eLearning.

Week 13: An application of multiple regression and course review

Learning objectives for this unit:

- This class will allow students to practice using the tools they have learned in an application of statistical modeling to business.
- We will review the course in the second session.

Review Session for TEST II: A review session will be provided at the regular class meeting time on Thursday, April 30.

TEST II: The test will be held at Testing Center at the first floor of the Synergy Park North 2 building --SPN2-- (3020 Waterview Parkway) on Tuesday, May 5 in any 2 hour 30 minutes time period from 8:30am to 6pm. Note that the exam location is NOT the regular classroom. Registration for seat reservation closes 48 hours prior to exam date!

TEST II covers Modules II and III. To prepare for Test II, review all your individual assignments, review lecture notes, and solve practice problems.

UT DALLAS SYLLABUS POLICIES AND PROCEDURES

The information contained in the following link constitutes the University's policies and procedures segment of the course syllabus.

Please go to UT Dallas Syllabus Policies webpage for these policies.

The descriptions and timelines contained in this syllabus are subject to change at the discretion of the Professor. Please check the course eLearning website regularly for following the announcements during the semester regarding such possible changes.